

Task-04 :Analyze and visualize sentiment patterns in social media data to understand public opinion and attitudes towards specific topics or brands.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from textblob import TextBlob
data=pd.read_csv('/content/online shopping.csv')
data
```

	Unnamed: 0	CustomerID	Gender	Location	Tenure_Months	Transaction_ID	Transaction_Date	Product_SKU	Product_Description	Price
0	0	17850.0	M	Chicago	12.0	16679.0	1/1/2019	GGOENEBJ079499	Nest Learning Thermostat 3rd Gen-USA - Stainle...	
1	1	17850.0	M	Chicago	12.0	16680.0	1/1/2019	GGOENEBJ079499	Nest Learning Thermostat 3rd Gen-USA - Stainle...	
2	2	17850.0	M	Chicago	12.0	16696.0	1/1/2019	GGOFNFRQ078999	Nest Cam Outdoor	

```
data= data.dropna(subset=['Product_Description'])
```

```
data['Clean_Text'] = data['Product_Description'].str.lower()
```

/tmp/ipython-input-1381883318.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Clean_Text'] = data['Product_Description'].str.lower()

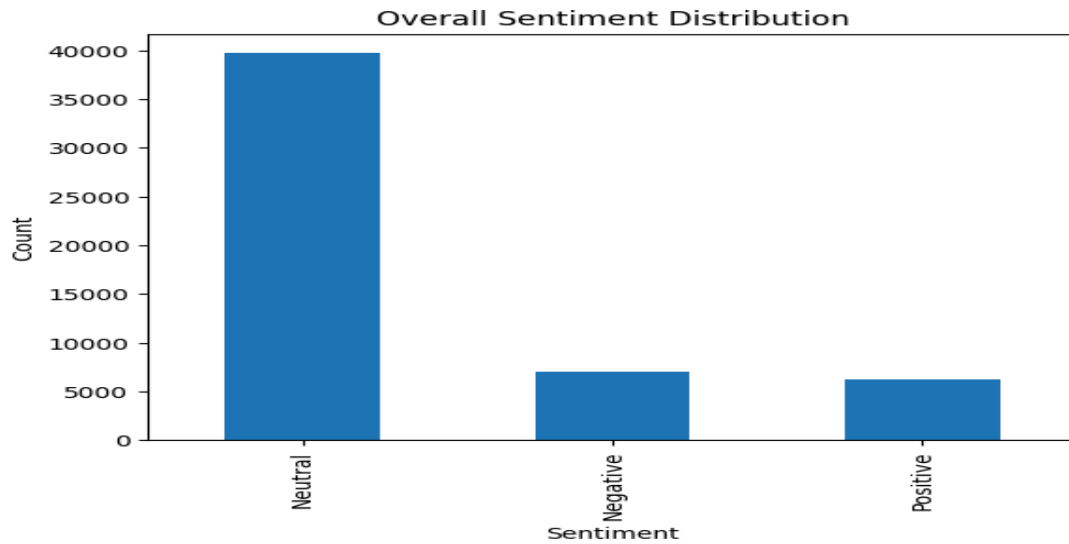
```
def get_sentiment(text):
    score = TextBlob(text).sentiment.polarity
    if score > 0:
        return 'Positive'
    elif score < 0:
        return 'Negative'
    else:
        return 'Neutral'
```

```
data['Sentiment'] = data['Clean_Text'].apply(get_sentiment)
```

... /tmp/ipython-input-2181837873.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

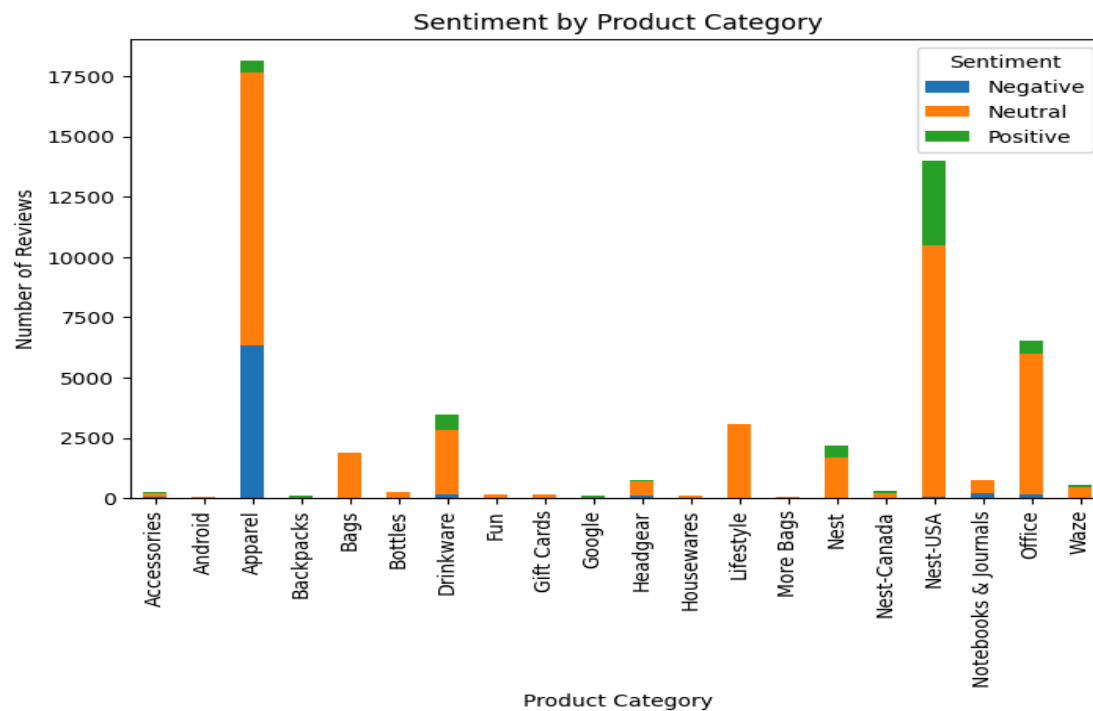
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Sentiment'] = data['Clean_Text'].apply(get_sentiment)

```
data['Sentiment'].value_counts().plot(kind='bar')
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.title("Overall Sentiment Distribution")
plt.show()
```



```
sentiment_category = pd.crosstab(data['Product_Category'], data['Sentiment'])

sentiment_category.plot(kind='bar', stacked=True, figsize=(8,5))
plt.xlabel("Product Category")
plt.ylabel("Number of Reviews")
plt.title("Sentiment by Product Category")
plt.show()
```



```
data.groupby('Sentiment')['Online_Spend'].mean().plot(kind='bar')
plt.xlabel("Sentiment")
plt.ylabel("Average Online Spend")
plt.title("Average Online Spend by Sentiment")
plt.show()
```

